

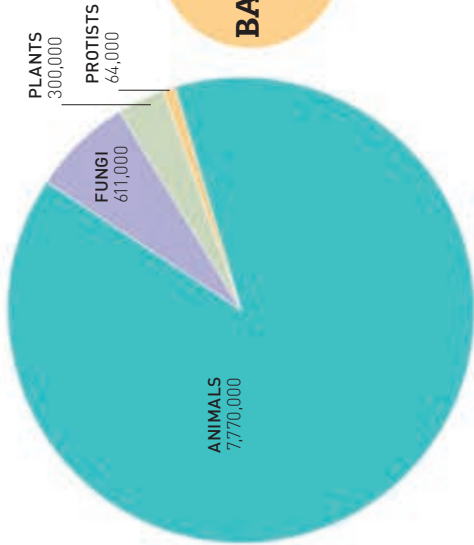


Birds

BRANCHING OUT

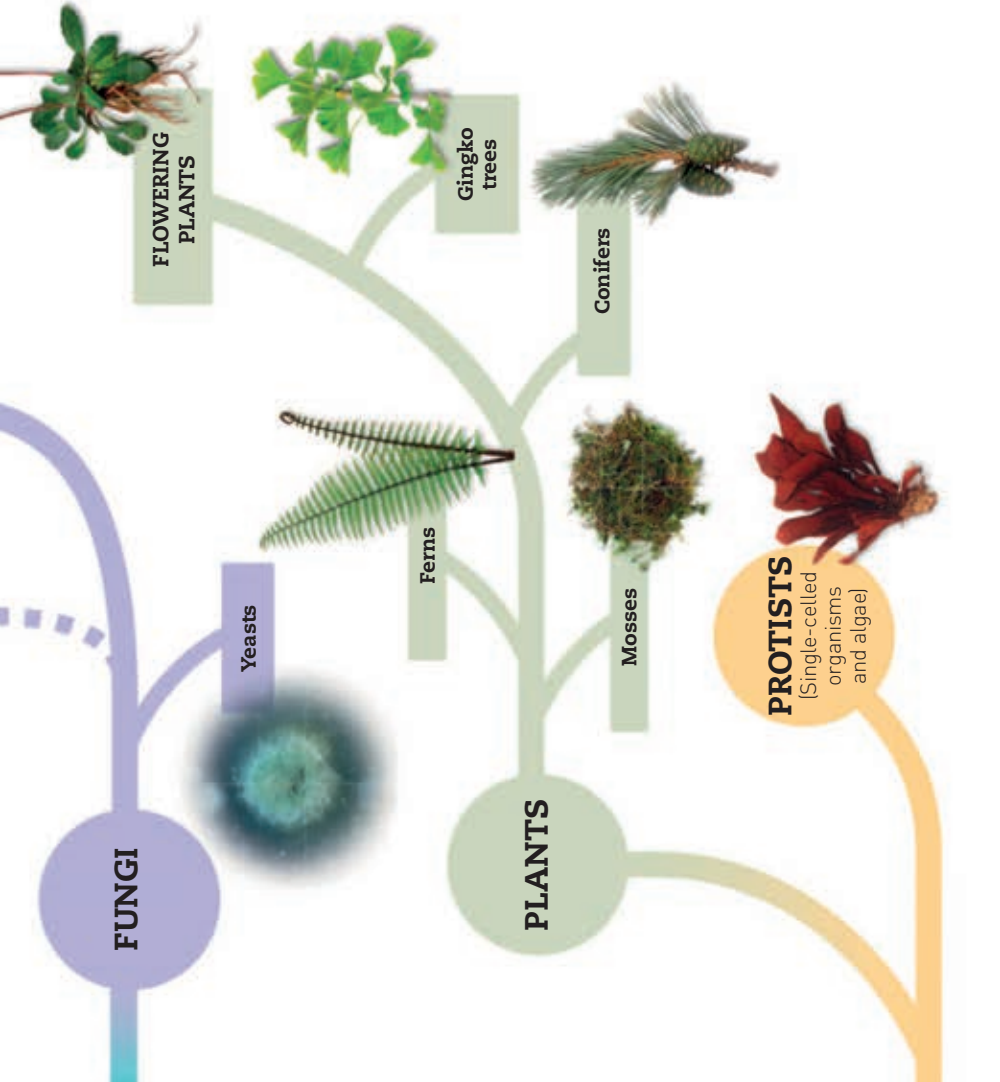
For well over two centuries, scientists have been using diagrams similar to this one to explain how life evolved. By following the “branches”, we can trace the relationships of the main groups of animals, plants, and fungi to the earliest types of life. Only living species are shown here. Extinct animals, such as dinosaurs, are not included.

ESTIMATED NUMBER OF SPECIES



HOW MANY SPECIES

There are far more species of animals than there are of plants, fungi, and protists added together. No one could possibly count up all the bacteria and archaea because there are simply too many millions of them.



Tree of life

The first living things that appeared on Earth, billions of years ago, were tiny organisms made of just one cell. This “tree” shows how such simple beginnings led to the development of the wonderful variety of life we know today.

CLASSIFICATION OF LIFE

Starting with the kingdoms, all living things are arranged, or classified, into further groups according to how they are related. This works in stages: the group called a phylum is

divided into classes, classes are split into orders, and so on. Shown below is how a tiger (scientific name *Panthera tigris*) is classified.

KINGDOM	PHYLUM	CLASS	ORDER	FAMILY	GENUS	SPECIES
1 ANIMALS Multi-celled living things that feed on other organisms. Most are able to move around.	2 CHORDATES Animals with a rod-like structure in their bodies. Includes vertebrates, which have a backbone.	3 MAMMALS Warm-blooded vertebrates with hair whose females feed their young on milk.	4 CARNIVORES Mostly mammals that hunt other animals for food and have special teeth for cutting through meat.	5 CATS Agile, specialized hunters. Many have sheathed toes into which their claws can disappear.	6 BIG CATS (PANTHERA) The largest and most powerful members of the cat family, including lions, tigers, and leopards.	7 TIGER (TIGRIS) The largest and heaviest of the forest-dwelling big cats, with distinctively striped fur.

THE SIX KINGDOMS

The tree of life is divided into six main branches, which scientists call kingdoms. Three kingdoms are mainly made up of tiny single-celled organisms. The others are fungi, plants, and animals.

ARCHAEA
Early life, made of one cell. Can live in extreme habitats, such as those that are very hot or salty.

BACTERIA
Cells similar to archaea but not suited to their extreme habitats. Many are vital to life; others cause diseases.

PROTISTS
Single cells with a nucleus. Some make their own food; others must feed on other organisms.

PLANTS
From tiny mosses to big trees, all plants use sunlight to make food and release oxygen into the air.

FUNGI
Some are single-celled, but the best known are mushrooms. They break down organic matter.

ANIMALS
These eat other organisms. Most use their senses for finding food. Some have backbones.

